

Synthetic MRPL Procurement Delegation of Financial Powers

Emergency Procurement and Single-Vendor Controls

FICTIONAL SYNTHETIC KNOWLEDGE-BASE DOCUMENT — FOR AI SYSTEM TESTING ONLY

This document is intentionally synthetic. It is not an authentic MRPL, PSU, procurement authority, government order, or contractual instrument.

Document ID	MRPL-DoFP-SYN-03
Revision	1.0
Status	Controlled synthetic test artifact
Purpose	RAG, routing, policy-verification, and document-generation evaluation
Primary topic	Emergency procurement from a single vendor

Prepared as a synthetic internal-style source for a sovereign, on-premise agentic AI workbench evaluation.

1. Document Control and Purpose

This document is a fictional procurement-policy artifact created for testing an on-premise agentic AI workbench. Its purpose is to provide a realistic, internally styled source against which an AI system can perform policy retrieval, rule interpretation, threshold comparison, approval identification, and deliverable generation. The document is deliberately separated from engineering, safety, and human-resources material so that a retrieval system must identify the correct source based on the subject of the request.

The central policy rule is intentionally precise. Within this synthetic dataset, emergency procurement of replacement equipment or shell plates exceeding INR 50,00,000 from a single vendor without an open public tender is strictly prohibited unless accompanied by a formal justification note. Final approval must be granted directly by the Unit Head. These statements are the controlling test facts for procurement-policy queries.

Explanatory sections in this document provide context for retrieval and reasoning tests. They do not create additional monetary limits, exceptions, approval authorities, or procurement routes beyond the explicit synthetic rule stated in the policy section. Where a user asks a question outside the facts represented here, the correct agent behavior is to identify the information gap rather than invent a policy.

2. Scope

The scope covers emergency procurement involving replacement equipment or shell plates where a single vendor is being considered without an open public tender. The document focuses on the financial threshold, the requirement for a formal justification note when the threshold is exceeded, and the requirement that final approval be granted directly by the Unit Head.

The scope does not define ordinary competitive procurement, routine purchase orders, employee matters, engineering fitness-for-service decisions, inspection limits, or plant-safety escalation. Those topics belong to other documents in the synthetic knowledge base. This separation is intentional and should be preserved during indexing and retrieval.

- Emergency procurement involving replacement equipment or shell plates.
- Single-vendor procurement without an open public tender.
- The synthetic INR 50,00,000 threshold.
- Formal justification where the stated threshold condition applies.
- Direct final approval by the Unit Head.
- Evidence and provenance needed for an AI-generated verification note.

3. Definitions

3.1 Emergency procurement

For this synthetic document, emergency procurement means a procurement action presented as urgent and handled under the emergency-procurement pathway rather than an open public tender. The document does not establish a separate test for what makes an event an emergency; the evaluator should therefore treat the user's characterization of a transaction as emergency procurement as an input to the policy-verification task.

3.2 Single vendor

Single vendor refers to a procurement arrangement in which the contemplated purchase is being sourced from one identified supplier rather than through an open public tender. The number of vendors and the procurement route are relevant because the controlling rule specifically addresses a single-vendor emergency purchase without an open public tender.

3.3 Formal justification note

A formal justification note is the documented explanation required by the synthetic policy when the specified threshold condition is triggered. The note should explain why the emergency procurement route is being used and provide enough information for the approving authority to understand the decision. This document does not prescribe a separate template or list of mandatory attachments, so an AI agent should not fabricate such requirements as though they were policy facts.

3.4 Unit Head

The Unit Head is the approval authority explicitly identified by this synthetic document for final approval in the covered emergency single-vendor case. The phrase final approval should not be weakened into a recommendation, endorsement, or routing step: within this test dataset, final approval is assigned directly to the Unit Head.

4. Governing Emergency Procurement Rule

The following is the principal retrieval target for this document.

Synthetic DoFP Rule
Any emergency procurement of replacement equipment or shell plates exceeding INR 50,00,000 from a single vendor, without open public tender, is strictly prohibited unless accompanied by a formal justification note. Final approval must be granted directly by the Unit Head.

The wording above should be treated as the authoritative synthetic rule for the evaluation. The threshold is INR 50,00,000. The procurement must involve replacement equipment or shell plates, a single vendor, and the absence of an open public tender. When the procurement exceeds the threshold, the policy requires a formal justification note before the transaction can proceed under the stated exception. Final approval remains with the Unit Head.

The rule is intentionally constructed so that an agent must reason over several conditions rather than match only a currency value. A query that mentions INR 68,00,000 but does not describe the procurement as emergency, single-vendor, and without open public tender would require the agent to identify which conditions are established and which remain unverified. In the benchmark query, all relevant conditions are supplied explicitly.

5. Threshold Interpretation

The monetary threshold in this synthetic document is expressed as INR 50,00,000, equivalent to fifty lakh rupees. The benchmark transaction is INR 68,00,000, equivalent to sixty-eight lakh rupees. The benchmark amount is therefore greater than the stated threshold by INR 18,00,000.

This comparison is a straightforward policy check. The agent should not confuse the threshold with an approval amount, a budget allocation, a tender value, or a purchase-order ceiling. In this synthetic source, the threshold determines whether the special justification condition is triggered for the described emergency single-vendor procurement.

Value	Amount	Interpretation
Policy threshold	INR 50,00,000	Reference threshold in the synthetic DoFP rule
Benchmark quote	INR 68,00,000	Exceeds the synthetic threshold
Difference	INR 18,00,000	Amount by which the benchmark exceeds the threshold

6. Conditions That Must Be Considered Together

The policy should be interpreted as a set of connected conditions. An agent should first identify the transaction type, then identify the procurement route, then compare the amount with the threshold, and finally determine the documentation and approval consequence.

- Transaction type: replacement equipment or shell plates.
- Urgency: the transaction is described as emergency procurement.
- Vendor structure: the purchase is from a single vendor.
- Tender route: the purchase is without an open public tender.
- Financial condition: the amount exceeds INR 50,00,000.
- Documentation consequence: a formal justification note is required by the synthetic rule.
- Approval consequence: final approval must be granted directly by the Unit Head.

7. Benchmark Case: INR 68,00,000 Single-Vendor Quote

The benchmark query presents an emergency quote from L&T; Heavy Engineering for INR 68,00,000 for replacement plates and asks the agent to verify whether the procurement is allowed and identify the required approval. For this synthetic benchmark, the named vendor is simply part of the test input; the procurement rule does not establish any vendor-specific exception.

The quote amount exceeds the INR 50,00,000 threshold. The transaction is described as emergency procurement, involves replacement plates, and is presented as a single-vendor purchase without an open public tender. The controlling rule therefore requires a formal justification note. The final approval must be granted directly by the Unit Head.

The correct policy-oriented conclusion is not that the purchase is automatically approved. Instead, the transaction falls under the restricted condition described by the rule and can proceed only with the formal justification note required by the synthetic exception, followed by direct final approval by the Unit Head. A generated note should use cautious policy language such as requires justification and Unit Head final approval rather than inventing an unconditional authorization.

8. Formal Justification Note

The justification note is the central documentary control introduced by the synthetic rule. It should establish the factual circumstances supporting the emergency route and explain why the proposed single-vendor procurement is being considered without an open public tender. The source does not prescribe a mandatory length, signatory list, annexure set, or commercial-evaluation format; therefore those details should not be represented as requirements derived from this document.

For an AI-generated draft, the note should identify the procurement subject, the quoted amount, the single-vendor nature of the transaction, the absence of an open public tender, and the fact that the amount exceeds the stated threshold. It should then state that the synthetic DoFP rule requires a formal justification note and direct final approval by the Unit Head.

The note should preserve provenance. If the quote amount is supplied by the user and the threshold comes from the knowledge base, the document can distinguish the user-provided transaction fact from the retrieved policy fact. This is useful in a sovereign AI workbench because an evaluator can inspect whether the policy was actually retrieved rather than generated from model memory.

9. Approval Authority

The synthetic policy assigns final approval directly to the Unit Head. This is the most important authority statement in the document. A routing workflow may involve preparation, review, or administrative handling, but the source does not define those additional roles. An AI agent should therefore avoid

adding a Procurement Committee, Finance Director, Plant Manager, Board, or other authority unless another source explicitly supplies that information.

For the benchmark query, the final approval field in a generated note should identify the Unit Head. If the user asks who must approve, the answer should be direct and grounded in this document. If the user asks for an approval hierarchy beyond the Unit Head, the agent should state that the supplied synthetic DoFP does not define additional levels.

10. Procurement Documentation and Evidence

A reliable policy-verification workflow should retain the evidence used to reach the conclusion. For the benchmark case, useful evidence includes the quoted amount of INR 68,00,000, the description of replacement plates, the single-vendor arrangement, the emergency nature of the procurement, and the relevant DoFP passage defining the INR 50,00,000 threshold and approval authority.

The evidence should be represented in the generated note without overstating facts. For example, if the user supplies the vendor name, the note may identify it as the quoted vendor. The policy source should be cited as the synthetic MRPL Procurement DoFP document. The agent should not imply that the synthetic document itself proves the commercial reason for selecting the vendor; it only provides the policy condition used for verification.

- Quote or commercial input supplied for the transaction.
- Procurement description and item category.
- Single-vendor status.
- Tender status as represented by the transaction record.
- Policy threshold and exception requirement.
- Required final approval authority.
- Source document identifier and revision.

11. Decision Matrix

Condition	Benchmark status	Policy effect
Emergency procurement	Yes	Covered by the emergency-procurement rule
Replacement equipment / shell plates	Replacement plates	Covered item category
Single vendor	Yes	Single-vendor condition met
Open public tender	No	Non-tender condition met
Amount over INR 50,00,000	INR 68,00,000	Threshold condition met
Formal justification note	Required	Must accompany the procurement under the synthetic rule
Final approval	Unit Head	Direct final approval required

12. Policy Interpretation Workflow

A policy-verification agent should follow a repeatable sequence. It should identify the user's transaction, determine which policy document governs the request, retrieve the relevant passage, extract the threshold and conditions, compare the transaction facts with those conditions, and produce a conclusion

with provenance. The sequence is intentionally different from an engineering calculation workflow. The agent should not activate a coding sandbox merely because a monetary comparison can be performed arithmetically. Likewise, it should not activate OCR or vision tools when the request is entirely textual. The first stage is intent recognition. A query asking whether a single-vendor emergency quote is allowed is a procurement-policy question. The router should therefore prioritize the RAG system and the document-generation tool if the user asks for a Word note. The second stage is evidence retrieval. The agent should locate the procurement DoFP document rather than searching every source equally. The third stage is condition extraction. The amount, procurement category, vendor arrangement, and tender status should be identified. The fourth stage is policy comparison. The amount is compared with the stated INR 50,00,000 threshold. The fifth stage is consequence identification: the formal justification note and Unit Head approval are retrieved from the same rule. The final stage is deliverable generation, where the conclusion is written in an auditable form.

13. RAG Retrieval Design

This document is intentionally structured to reward precise retrieval. Queries may use terms such as emergency procurement, single vendor, replacement plates, financial powers, threshold, justification note, or Unit Head. A good semantic retriever should locate the governing rule even if the wording of the user differs slightly. For example, a user may ask about a sixty-eight-lakh single-source emergency purchase rather than repeating the exact phrase single vendor. The agent should still connect the request to the relevant policy passage when the other conditions align. Retrieval quality should be evaluated separately from generation quality. If the correct source is never retrieved, a polished final note can still be wrong. The benchmark should therefore log the documents selected by the RAG component. For Query 2, the expected relevant source is this procurement DoFP document. Engineering material limits, corrosion rates, OISD safety rules, and HR leave information are not required for the policy verification. The HR document is particularly useful as noise because it is also an internal corporate policy and can therefore appear superficially relevant to a generic policy query. Chunking should preserve the threshold and its conditions together. A chunk that contains only INR 50,00,000 without the surrounding procurement context is less useful than a chunk that contains the threshold, the single-vendor condition, the tender condition, the justification requirement, and the Unit Head approval requirement. Metadata such as document ID, revision, and document type can supplement the text but should not replace source content.

14. Noise Filtering and Source Boundaries

The synthetic knowledge base contains four other documents. The API-style inspection code concerns material retirement limits. The CDU-1 SOP concerns corrosion rates and remaining operational life. The OISD safety guideline concerns a Level 3 Fitness-For-Service assessment when remaining life is strictly below six years. The HR leave policy concerns employee block leave. None of those subjects is necessary to verify the benchmark procurement quote. Noise filtering is therefore an important part of the test. A broad query such as check our policies could potentially retrieve several internal-looking documents. A robust router should use the user's actual task to select procurement policy as the dominant intent. The RAG layer should then return the DoFP source because it contains the financial and approval rule. If unrelated documents appear in intermediate retrieval results, the reranker or agent should exclude them before forming the final answer. The system should also avoid cross-document contamination. For example, the existence of a Unit Head in the procurement document should not cause the agent to claim that the Unit Head is an approval authority for engineering or safety decisions. Similarly, the OISD document should not be cited in a procurement note merely because both are internal-style documents. The benchmark rewards disciplined source selection rather than the maximum number of retrieved passages.

15. Generated Word Note Requirements

When the user requests a formal Word note, the generated deliverable should be concise enough for an executive reviewer while still showing the evidence behind the conclusion. A useful structure is a subject line, transaction summary, policy reference, condition-by-condition verification, compliance conclusion,

required action, approval authority, and source note. The transaction summary should identify the replacement plates, the quoted amount of INR 68,00,000, the single-vendor nature of the quote, and the emergency procurement context. The policy section should state the synthetic DoFP threshold and its consequence. The verification section should show that INR 68,00,000 exceeds INR 50,00,000. The conclusion should state that a formal justification note is required and that final approval must be granted directly by the Unit Head. The note should not say that the procurement is unconditionally permitted. The more accurate synthetic conclusion is that the transaction is subject to the policy condition and may proceed only when the required formal justification note accompanies the procurement and the Unit Head provides final approval. If the user asks for a note that simply says approved, the agent should not invent approval that has not been granted.

16. Provenance and Auditability

An agentic workbench should make it possible to determine where every important statement originated. In this benchmark, the quote amount and vendor are user-supplied transaction facts, while the INR 50,00,000 threshold, formal justification requirement, and Unit Head approval requirement come from the synthetic DoFP source. The difference between these categories matters because an auditor may need to verify policy language independently of the transaction record. A generated note can include a source-reference section identifying the document as MRPL-DoFP-SYN-03, Revision 1.0. The note may also identify the relevant section as the Governing Emergency Procurement Rule. This makes it easier to inspect the RAG result and reproduce the decision. Provenance is especially valuable in a confidential on-premise deployment because the objective is not merely to obtain a fluent answer but to demonstrate that the assistant works from controlled internal knowledge without sending the underlying material to external services.

17. Edge Cases for Evaluation

Several edge cases can be used to evaluate the policy agent. If a quote is exactly INR 50,00,000, the wording of the source says exceeding INR 50,00,000, so the source does not say that the threshold condition is triggered at exactly INR 50,00,000. The agent should preserve that distinction rather than silently changing exceeding to equal to or greater than. If the amount is INR 49,00,000, the transaction is below the stated threshold, but the document does not provide enough information to determine every other procurement requirement. The agent should therefore avoid claiming that the entire procurement is universally permitted. If the transaction is above INR 50,00,000 but an open public tender is used, the exact synthetic rule is not directly triggered because it specifically describes a single-vendor purchase without an open public tender. Again, the correct behavior is to state what the source establishes and not invent a separate rule. If the purchase concerns an item other than replacement equipment or shell plates, the applicability of this particular rule may be uncertain. The agent should flag that uncertainty rather than broaden the source's scope without evidence. If the user asks for an approval authority and provides no transaction details, the agent can state that for the covered emergency single-vendor condition, final approval is assigned directly to the Unit Head. It should not assume that every procurement at every value requires that authority because the source does not establish such a universal rule.

18. Security and Confidentiality Role in the Synthetic Test

The document is designed for a sovereign AI workbench in which sensitive company information is processed locally. Procurement data can include vendor names, quotations, commercial values, negotiations, and approval correspondence. In a real environment, such information may be confidential and should remain within the organization's controlled infrastructure. The synthetic document itself contains no real confidential transaction, but its structure imitates the type of policy source an on-premise system would index. The evaluation should therefore test more than answer accuracy. It should also demonstrate that the RAG component can operate against a local knowledge base, that generated documents can be produced locally, and that no external network call is necessary to answer the benchmark query. Logs or a visible network monitor can be used by the demonstrator to show the sovereign-processing claim. This source should never be used to imply that a particular real-world

company or vendor follows these synthetic rules. The names and policy framing are test artifacts. The intended evaluation is of the AI system's retrieval, routing, reasoning, and document-generation behavior.

19. Regression Testing

Once the five-document knowledge base is loaded, this document can serve as a stable regression target. A future change to the embedding model, reranker, router, or language model should continue to retrieve the procurement document for the benchmark query. If the agent begins selecting the HR leave policy because both are corporate-policy documents, the retrieval test has exposed a precision regression. A second regression test is numerical interpretation. The amount INR 68,00,000 should be recognized as greater than INR 50,00,000. A third test is authority extraction: the Unit Head should be identified as the final approval authority. A fourth test is deliverable generation: the Word note should contain the required justification and approval statements and should not cite unrelated engineering or HR documents. The benchmark can be repeated after changing chunk size, embedding models, retrieval top-k, reranking settings, or the model selected by the task router. Stable source documents make it possible to attribute changes in behavior to system configuration rather than changes in the underlying knowledge.

20. Acceptance Checklist

An evaluator can use the following checklist to determine whether the procurement-policy workflow is functioning correctly.

- The procurement query is classified as a policy-verification task.
- The RAG system retrieves the MRPL procurement DoFP source.
- The engineering, OISD, and HR documents are not used as governing evidence.
- The transaction is recognized as emergency procurement.
- The transaction is recognized as involving replacement plates.
- The single-vendor condition is recognized.
- The absence of an open public tender is recognized.
- INR 68,00,000 is compared against the INR 50,00,000 threshold.
- The amount is recognized as exceeding the threshold by INR 18,00,000.
- A formal justification note is identified as required.
- The Unit Head is identified as the direct final approval authority.
- The generated Word note distinguishes transaction facts from policy facts.
- The final note does not claim unconditional approval.
- The source document identity and synthetic status can be traced.

21. Example Executive Approval Note Content

Subject: Verification of Emergency Single-Vendor Procurement for Replacement Plates

A single-vendor emergency quotation has been received for replacement plates at a quoted value of INR 68,00,000. The applicable synthetic MRPL Delegation of Financial Powers rule states that emergency procurement of replacement equipment or shell plates exceeding INR 50,00,000 from a single vendor without an open public tender is strictly prohibited unless accompanied by a formal justification note. The same rule requires final approval to be granted directly by the Unit Head.

The quoted value of INR 68,00,000 exceeds the stated INR 50,00,000 threshold by INR 18,00,000. On the facts provided, the transaction falls within the condition addressed by the synthetic rule. Accordingly, a formal justification note is required, and final approval must be obtained directly from the Unit Head before proceeding under the stated emergency-procurement pathway.

This example is included to demonstrate the expected semantic content of a generated deliverable. It is not itself an approval and must not be interpreted as one. The source is a fictional test artifact.

22. Limitations and Controlled Use

This document is synthetic and exists solely to support testing of a local AI knowledge base and agentic workflow. It is not an authentic MRPL document, official Delegation of Financial Powers, government notification, procurement manual, or legal instrument. It should not be used to make actual procurement decisions.

The document intentionally contains only the policy facts needed for the benchmark. It does not attempt to describe the complete procurement framework of any organization. It does not define tendering procedures, vendor due diligence, tax treatment, purchase-order processes, budget availability, commercial negotiations, contract law, or other controls unless explicitly stated. Those omissions are deliberate because the goal is to test whether an AI system respects the boundary of its evidence.

In a real deployment, the synthetic source should be replaced by approved and controlled organizational documents. The agent should retrieve the current revision of the actual policy and preserve the organization's document-control requirements. Until then, every statement in this file should be treated as test data only.

23. Revision History

Revision	Description	Status
1.0	Initial synthetic procurement-policy artifact for AI evaluation	Controlled test artifact